

PROJECT BRIEF

1. THE QUESTION

Which neighborhoods in Yaoundé contain the largest populations living in low-lying areas within 200 metres of a waterway?

2. WHY IT MATTERS

Yaoundé's dense urban development, varied terrain and numerous waterways create places where populations may be more exposed to flooding. Identifying where large populations overlap with relatively low-lying terrain and proximity to waterways can help urban planners and disaster-risk stakeholders identify priority areas for further investigation. This project is a screening analysis, not a flood prediction model.

3. THE DATA I NEED

Dataset	Purpose
Yaoundé city boundary + mapped waterways	Define the study area and create the 200 m waterway buffer.
SRTM elevation (~100 m)	Identify the lowest 20% of terrain within Yaoundé as the project's relative low-lying zone.
WorldPop Cameroon population, 2020 (~100 m)	Estimate the population contained in the potential exposure zone.
Cameroon administrative boundaries	Aggregate and rank results by neighborhood where reliable boundaries exist; otherwise use the most appropriate administrative unit.

4. WHERE EACH DATASET COMES FROM

TUMI Datahub — Base Dataset Yaoundé, Cameroon — Yaoundé boundary and OSM-derived mapped waterways. Updated 27 April 2025; GeoJSON, WGS 84.

<https://hub.tumidata.org/dataset/f5e30a84-cba8-4cdf-99be-67148fcb9a9f>

WorldPop — SRTM elevation, Cameroon — SRTM-derived elevation at 3 arc-seconds (approximately 100 m); GeoTIFF, WGS 84.

<https://hub.worldpop.org/geodata/summary?id=23306>

WorldPop — Population Counts, Cameroon, 2020 — Estimated population per grid cell at 3 arc-seconds (approximately 100 m); GeoTIFF, WGS 84.

<https://hub.worldpop.org/geodata/summary?id=6347>

UN SALB — Cameroon administrative units — Validated administrative polygon data for Cameroon, available in GeoJSON, Shapefile and other GIS formats.

<https://salb.un.org/en/data/cmr>

5. WHAT I WILL BUILD

A Yaoundé Population Exposure to Low-Lying Areas Near Waterways Map, plus a ranked table showing the areas with the largest estimated populations inside the potential exposure zones. The workflow will clip the data to Yaoundé, derive the lowest 20% of terrain, buffer mapped waterways by 200 m, intersect the two zones, overlay WorldPop population, and aggregate the results by the selected geographic unit. The final product will be a reproducible screening tool for identifying priority areas for further flood-exposure and urban-resilience investigation.